VIT

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

Office of the Controller of Examinations

Transcript

NAME: SUBHRANSU SEKHAR BHATTACHARJEE **REGISTRATION NUMBER: 18BEC0576**

PROGRAMME: BACHELOR OF TECHNOLOGY, ELECTRONICS AND COMMUNICATION ENGINEERING **BATCH: 2018**

Course Code	Course Title	Credits	Grade	Marks	Paper Type
MEE1002	ENGINEERING MECHANICS	3	A	81	Theory
CSE1001	PROBLEM SOLVING AND PROGRAMMING	4	S	97	Theory-Lab
ECE1001	FUNDAMENTALS OF ELECTRIC CIRCUITS	3	S	90	Theory-Lab
MAT1011	CALCULUS FOR ENGINEERS	4	S	93	Theory-Lab
PHY1701	ENGINEERING PHYSICS	4	A	81	Theory-Lab-Project
ECE1004	SIGNALS AND SYSTEMS	3	S	92	Theory-Lab
EEE1007	NEURAL NETWORKS AND FUZZY LOGIC	3	A	84	Theory-Lab-Project
MAT1004	DISCRETE MATHEMATICS AND GRAPH THEORY	4	A	81	Theory
MAT2001	STATISTICS FOR ENGINEERS	4	S	94	Theory-Lab
MAT2002	DIFFERENTIAL AND DIFFERENCE EQUATIONS	4	S	97	Theory-Lab
ECE1002	SEMICONDUCTOR DEVICES AND CIRCUITS	4	A	84	Theory-Lab-Project
ECE2002	ANALOG ELECTRONIC CIRCUITS	4	A	80	Theory-Lab-Project
ECE1007	OPTO ELECTRONICS	3	B	71	Theory-Lab-Project
MAT2100	APPLIED LINEAR ALGEBRA	4	S	91	Theory-Lab
ECE1003	ELECTROMAGNETIC FIELD THEORY	3	A	82	Theory
ECE2004	TRANSMISSION LINES AND WAVEGUIDES	3	A	84	Theory-Lab
ECE2002	NETWORK THEORY	3	S	94	Theory
ECE2006	DIGITAL SIGNAL PROCESSING	4	A	87	Theory-Lab-Project
PHY1711	INTRODUCTION TO INNOVATIVE PROJECTS	2	B	76	Lab-Project
PHY1322	MATERIAL SCIENCE	4	A	81	Theory-Lab
ECE2005	PROBABILITY THEORY AND RANDOM PROCESSES	3	S	94	Theory-Lab
CSE2003	DATA STRUCTURES AND ALGORITHMS	4	A	80	Theory-Lab-Project
CSE1002	OBJECT ORIENTED PROGRAMMING	3	A	83	Theory-Lab-Project

----------END OF STATEMENT----------

VELLORE INSTITUTE OF TECHNOLOGY (VIT) VELLORE-14 CONTROLLER OF EXAMINATIONS

VIT

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

Total unit of credits completed: 80

CGPA: 9.34

Medium of instruction is English.

Grading System

A letter grade on a ten point scale is awarded to a student in a course based on the total marks obtained from Continuous Assessment Tests and Final Assessment Tests.

Letter Grade	S	A	B	C	D	E	F	N
Grade Point	10	9	8	7	6	5	0	0
F	Failed in a course							
N	Absent/Debarred/Course incomplete							
P	Passed							
U	Audited							

Grades S, A, B, C, D, E, P and U indicate successful completion of course.

CGPA calculation

CGPA is obtained by dividing the sum of product of Grade Points (GP_i) earned and Credits (C_i) of a registered course for 'n' number of courses by total number of credits registered.

$$\frac{\sum_{i=1}^{n} GP_i \times C_i}{\sum_{i=1}^{n} C_i}$$

VELLORE INSTITUTE OF TECHNOLOGY (VIT) VELLORE-14 CONTROLLER OF EXAMINATIONS

Verified by:

Processed by:

Issued on: 06 July 2020

Controller of Examinations

VIT®
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

VIT / AR / 2020 / 2000009439 **Date: 06.11.2020**

MIGRATION CERTIFICATE

Certified that this Institution has no objection for the following student to join any other University for further studies.

Name of the Student	:	**MR. SUBHRANSU SEKHAR BHATTACHARJEE**
Registration No.	:	**18BEC0576**
Name of the Father / Guardian	:	**MR. S S BHATTACHARYA**
Date of Birth	:	**18.12.1998**
Course Undergone	:	**B.TECH. - ELECTRONICS AND COMMUNICATION ENGINEERING**
Date of Admission	:	**14.05.2018**
Date of Leaving	:	**06.11.2020**
Reason for Leaving	:	TRANSFERRED TO ANU
TC Issued Date / Sl.No.	:	**06.11.2020 / 2000009439**

The Character and Conduct of the Student during the course of study was:

Excellent

Vellore Institute of Technology ★ Vellore-632 014, Tamil Nadu, India ★ Record Section

REGISTRAR

Vellore - 632 014, Tamil Nadu, India; Phone: 91 - 416 - 2243091 (10 Lines) Fax: 91- 416 - 2243092
E-mail: registrar@vit.ac.in
www.vit.ac.in

VALID ONLY FOR ONLINE VIEWING AT https://www.myequals.net

Australian National University

Australian Higher Education Graduation Statement

1. the graduate

Name	Subhransu Bhattacharjee
Student number	7143478

2. the award

Name of award

Bachelor of Engineering With Honours

Detail

The Bachelor of Engineering (Honours), BEngHons, is an AQF Level 8 Bachelor Honours qualification taught in English and normally takes four years of full-time-equivalent study. The degree is based on a multidisciplinary, systems engineering philosophy that is designed to produce well rounded and multi-skilled engineers. The curriculum provides students with the domain knowledge relevant to their field and engenders in them a way-of-thinking that provides them with the confidence to be creative even when the problem they encounter may be unfamiliar. The degree structure requires the completion of a compulsory professional core into which the systems approach is built, as well as a technical specialisation. Students complete a supervised capstone individual final year project. Admission is normally based on secondary education academic results.

Features

The Bachelor of Engineering (Honours) offers technical specialisations (majors and minors) in a range of areas. These are indicated on the degree transcript. The degree allows for up to 36 units of university electives which may be additional engineering courses. The Bachelor of Engineering (Honours) offers candidates the opportunity for industry placements and overseas study experiences. Students are required to undertake 12 weeks of professional work experience in order to graduate.

Pathway to further study

Graduates may progress to an AQF level 9 qualification (Masters Degree) or AQF level 10 qualification (Doctoral Degree), subject to specific admission requirements.

Course accreditation

Graduates of the Bachelor of Engineering (Honours) will be eligible to apply for professional membership of Engineers Australia, Australia's professional engineering association. This may also entitle them to practice in countries covered in the joint recognition agreement, the Washington Accord.

3. awarding institution

The Australian National University is a research intensive education institute established by an Act of the Australian Parliament on 1 August 1946 and is listed as an Australian University on the Tertiary Education Quality and Standard Agency's National Register of Higher Education Providers. The University's Commonwealth Register of Institutions and Courses for Overseas Students number is 00120C.

For more information about The Australian National University visit www.anu.edu.au

The Australian Higher Education Graduation Statement is provided by Australian higher education institutions to graduating students on completion of the requirements for a particular higher education award. It provides a description of the nature, level, context and status of studies that were pursued by the individual named. Its purpose is to assist in both national and international recognition of Australian qualifications and to promote international mobility and professional recognition of graduates.

certification

date 13 December 2022

Scott Pearsall
Registrar, Student Administration
& Academic Services

VALID ONLY FOR ONLINE VIEWING AT https://www.myequals.net

VALID ONLY FOR ONLINE VIEWING AT https://www.myequals.net

Australian National University

Australian Higher Education Graduation Statement

4. graduate's academic achievements

Bachelor of Engineering With First Class Honours awarded 13 December 2022

Major in Mechatronic Systems
Minor in Electronic and Communication Systems
Minor in Mathematics

Course Code	Course Title	Units	Mark	Grade
2020	**STATUS**			
COMP 1730	Programming for Scientists	6		STE
ENGN 1211	Discovering Engineering	6		STE
ENGN 1215	Engineering Sciences	6		STE
ENGN 1217	Introduction to Mechanics	6		STE
ENGN 1218	Introduction to Electronics	6		STE
ENGN 2217	Mechanical Systems and Design	6		STE
ENGN 2218	Electronic Systems and Design	6		STE
ENGN 2228	Signals and Systems	6		STE
ENGN 2996	Unspecified 2000 Series Non-Engineering Elective	12		STE
ENGN 4537	Digital Signal Processing	6		STE
MATH 1005	Discrete Mathematical Models	6		STE
MATH 1013	Mathematics and Applications 1	6		STE
MATH 1014	Mathematics and Applications 2	6		STE
MATH 2901	Unspecified 2000 Level Maths Course	6		STE
PHYS 1101	Physics I	6		STE
2020	**SECOND SEMESTER**			
ENGN 2301	Engineering Design 3: Systems Approaches for Analysis	6	CRS	CRS
ENGN 3223	Control Systems	6	71	D
ENGN 4625	Power Systems and Power Electronics	6	72	D
POLS 1006	Introduction to International Relations: Contemporary Global Issues	6	CRS	CRS
2021	**FIRST SEMESTER**			
ENGN 3300	Engineering Design 4A: Systems Approaches for Management	6	76	D
ENGN 3331	System Dynamics	6	78	D
ENGN 4200	Individual Project	6	93	HD
ENGN 4528	Computer Vision	6	CRS	CRS
2021	**SECOND SEMESTER**			
ENGN 2222	Engineering Thermodynamics	6	75	D
ENGN 4200	Individual Project	6	93	HD
2022	**FIRST SEMESTER**			
ENGN 2219	Computer Systems & Organisation	6	76	D
ENGN 2300	Engineering Design 2: Systems Approaches for Design	6	82	HD
ENGN 4213	Digital Systems and Microprocessors	6	87	HD
2022	**WINTER SESSION**			
ENGN 3100	Practical Experience	0	WD	WD
2022	**SECOND SEMESTER**			
ENGN 3100	Practical Experience	0	CRS	CRS
ENGN 3301	Engineering Design 4B: Systems Approaches for Operations	6	82	HD

VALID ONLY FOR ONLINE VIEWING AT https://www.myequals.net

VALID ONLY FOR ONLINE VIEWING AT https://www.myequals.net

Australian National University

Australian Higher Education Graduation Statement

Course Code	Course Title	Units	Mark	Grade
ENGN 4100	Engineering Honours Result	0	H1	H1
ENGN 4627	Robotics	6	86	HD
MGMT 2100	Communication for Business	6	76	D

VALID ONLY FOR ONLINE VIEWING AT https://www.myequals.net

VALID ONLY FOR ONLINE VIEWING AT https://www.myequals.net

5. description of the australian higher education system

Introduction

The Australian higher education system consists of self-governing public and private universities and higher education institutions that award higher education qualifications.

The Australian Qualifications Framework

The Australian Qualifications Framework (AQF) is a single national, comprehensive system of qualifications offered by higher education institutions (including universities), vocational education and training institutions and secondary schools.

Australian Qualifications Framework

The AQF has 10 levels, each with defined criteria based on a taxonomy of learning outcomes. Higher education qualifications are placed between level 5 (the Diploma) and level 10 (the Doctoral Degree). The Bachelor Degree is at level 7. Each AQF qualification has a set of descriptors which define the type and complexity of knowledge, skills and application of the knowledge and skills that a graduate who has been awarded that qualification has attained, and the typical volume of learning associated with that qualification type. The full set of levels criteria and qualification type descriptors can be found by visiting www.aqf.edu.au.

The main AQF qualifications awarded by higher education institutions are Bachelor Degrees, Masters Degrees and Doctoral Degrees. There are also three qualifications at the sub-degree level: the Diploma, the Advanced Diploma and the Associate Degree. At the graduate level but below the Masters Degree are the Graduate Certificate and Graduate Diploma. The Undergraduate Certificate, which was added to the AQF in 2020, is a higher education qualification that is not allocated to a single AQF level, but recognises completion of a short course, equivalent to half a year's study (0.5 Equivalent Full Time Study Load) toward an existing qualification at AQF level 5, 6 or 7.

Level	Qualification Type	Summary
-	Senior Secondary Certificate of Education	Graduates with this qualification will have knowledge, skills and values for diverse pathways to further learning, work and effective participation in civic life
Level 1	Certificate I	Graduates at this level will have knowledge and skills for initial work, community involvement and/or further learning
Level 2	Certificate II	Graduates at this level will have knowledge and skills for work in a defined context and/or further learning
Level 3	Certificate III	Graduates at this level will have theoretical and practical knowledge and skills for work and/or further learning
Level 4	Certificate IV	Graduates at this level will have theoretical and practical knowledge and skills for specialised and/or skilled work and/or further learning
Level 5	Diploma	Graduates at this level will have specialised knowledge and skills for skilled and/or paraprofessional work and/or further learning
Level 6	Advanced Diploma Associate Degree	Graduates at this level will have broad knowledge and skills for paraprofessional/highly skilled work and/or further learning
Level 7	Bachelor Degree	Graduates at this level will have broad and coherent knowledge and skills for professional work and/or further learning
Level 5-7	Undergraduate Certificate	Graduates with this qualification will have foundational knowledge and skills for further study, professional upskilling, work and participation in lifelong learning
Level 8	Bachelor Honours Degree Graduate Certificate Graduate Diploma	Graduates at this level will have advanced knowledge and skills for professional highly skilled work and/or further learning
Level 9	Masters Degree	Graduates at this level will have specialised knowledge and skills for research, and/or professional practice and/or further learning
Level 10	Doctoral Degree	Graduates at this level will have systematic and critical understanding of a complex field of learning and specialised research skills for the advancement of learning and/or for professional practice

*As at the publication of this document, the Undergraduate Certificate qualification type must not be conferred after 30 June 2025. Undergraduate Certificates that have been conferred prior to this date will continue to be recognised as valid AQF qualifications.

VALID ONLY FOR ONLINE VIEWING AT https://www.myequals.net

VALID ONLY FOR ONLINE VIEWING AT https://www.myequals.net

Admission

Requirements for admission to particular awards are set by higher education institutions and provide a range of routes for entry and only admit those students considered to have potential to complete an award successfully. Admission of school leavers to undergraduate awards is typically on the basis of the level of achievement in Year 12 secondary education, although some institutions and awards also use interviews, portfolios or demonstrated interest or aptitude. Most institutions also provide alternative entry provisions via bridging or foundation programs for mature age students or other special provisions, such as recognition of prior learning from previous study. Admission to post-graduate awards is generally based on the level of achievement in previous higher education studies and in most cases, admission to PhD awards is based on high achievement in a research Masters Degree or in a Bachelor Degree with first class honours or second class honours division A.

Quality

Quality assurance and stringent approval requirements for higher education institutions ensure that Australia has an international reputation for high quality education.

The Tertiary Education Quality and Standards Agency (TEQSA) was established on 30 July 2011 as a new national regulator and quality assurance agency for higher education. TEQSA is an independent body with the powers to regulate university and non-university higher education institutions and monitor quality against standards. Its purpose is to protect student interests and the reputation of Australia's higher education sector through a proportionate, risk-reflective approach to quality assurance that supports diversity, innovation and excellence.

From 29 January 2012 TEQSA assumed responsibility for registering and re-registering providers and accrediting and re-accrediting awards for higher education providers that do not have authority to accredit their own awards. At the time of registration, re-registration, accreditation and/or re-accreditation, TEQSA evaluates the performance of a higher education provider against the Higher Education Standards Framework. The framework specifies the minimum standards that all higher education providers must meet and continue to meet in order to remain registered.

TEQSA also monitors provider risks and performance and identifies good practice to support continuous improvement within the sector, guide sectoral quality enhancement and inform policy and research.

The Higher Education Standards Panel, which is independent from TEQSA, is responsible for advising the Minister for Education and TEQSA on the development, monitoring and interpretation of the Higher Education Standards Framework.

All higher education institutions receiving Australian Government financial support must meet quality and accountability requirements that are set out in the Higher Education Support Act 2003. The Australian Government also uses a range of tools to measure and monitor the quality of outcomes, while the interests of international students are protected by the Education Services for Overseas Students Act 2000 and the Commonwealth Register of Institutions and Courses for Overseas Students (CRICOS), providing tuition assurance and ensuring that institutions listed on CRICOS meet defined minimum standards.

VALID ONLY FOR ONLINE VIEWING AT https://www.myequals.net

The Australian National University Australian Higher Education Graduation Statement

Explanation of grades and codes

Key to grades and codes					
HD	High Distinction	80 - 100	H1	First Class Honours	80 - 100
D	Distinction	70 - 79	H2A	Second Class Honours Division A	70 - 79
CR	Credit	60 - 69	H2B	Second Class Honours Division B	60 - 69
P	Pass	50 - 59	H3	Third Class Honours	50 - 59
CRS	Course Requirements Satisfied	ungraded pass	IP	In progress (continuing course)	
PS	Pass at supplementary exam	50	KU	Continuing course	
HLP	Higher Level Performance	higher ungraded pass	EE	Enrolled elsewhere	
N	Fail	0 - 49	EXE	Exemption external (no unit value)	
NCN	Not complete/Fail	not graded	EXI	Exemption internal (no unit value)	
WD	Withdrawn without failure	not graded	RC	Research continuing	
WN	Withdrawn with failure	not graded	STE	Status external (external credit)*	
WL	Withdrawn late without failure	not graded	STI	Status internal (internal credit)*	

This explanation is current at 1 July 2011. Subsequent amendments and further details of assessment notations are available at **drss.anu.edu.au**

*Results for internal status credited to the specified award are not recorded on this Graduation Statement. Please see the graduate's Transcript of Academic Record for these results. Results for any external status credited cannot be provided by ANU.

Unit values and equivalencies

The majority of courses offered at ANU are valued at 6 units. The normal maximum load in a semester or half-year is 24 units (normally 4 courses). Undertaking 18-24 units per semester is considered a full-time study load.

ANU considers 30 European Credit Transfer System (ECTS) credits as equivalent to 24 ANU units.

Course information

Details of the syllabus for individual courses listed on this Australian Higher Education Graduation Statement may be found in the relevant University Handbook for the year indicated. Alternatively, course information may be obtained from the relevant academic area.

Semesters and sessions

Coursework programs		Research programs	
First Semester	February - June	First Semester	1 January - 30 June
Second Semester	July - November	Second Semester	1 July - 31 December
Summer Session	January - March	Quarter 1	January - March
Autumn Session	April - June	Quarter 2	April - June
Winter Session	July - September	Quarter 3	July - September
Spring Session	October - December	Quarter 4	October - December

Transcript of Academic Record

The information contained within this Australian Higher Education Graduation Statement pertains to the specified award only. For a complete history of academic results achieved at ANU, or results for courses granted as internal status, please see the graduate's Transcript of Academic Record.

Glossary of terms

Terminology	Explanation
Award	A certificate, diploma, degree, associate degree, graduate certificate or graduate diploma, conferred by the University to a person upon the completion of an academic program leading to an award, for example a Bachelor of Arts. The award is represented by a testamur and is presented to a graduate upon graduating.
Course	A discrete entity of study in a subject, normally of one semester or session in duration, identified by an alpha-numeric code. For example, 'ENGN3224 Energy Systems Engineering' is one course. Three or four courses per semester or half-year normally constitutes full-time enrolment.
Grade	A classification of achievement based on the final mark. For example a grade of High Distinction (HD) indicates a final mark between 80 and 100.
Mark	A number indicating a student's performance in an assessment activity or a course. Raw marks, which may be scaled, become final marks after approval by the Delegated Authority. Marks range from 0 to 100.
Program	A structured sequence of study, undertaken in one or more Colleges of the University, normally leading to an award. Programs are controlled by one or, in the case of combined programs, two academic authorities.
Status	Students admitted to a program at the University, or transferring between programs at the University, may be granted status (credit) in the new program on the basis of previous tertiary studies.
Unit	This is an indicator of the value of a course within the total program. Most courses are valued at 6 units.

Document security

This document is printed using a number of special security features including watermarked paper, microtext printing and University hologram. To confirm this document is an original, hold the paper up to a light source to view a wavy line image throughout the paper. Additionally, the University hologram appears on each page of this document, is multicoloured, multilayered and includes the University crest. The absence of any of these security features may indicate that this document is not an original and should not be accepted as an official University document. Any questions regarding this document should be made to the Division of Registrar and Student Services at **drss.anu.edu.au**

VALID ONLY FOR ONLINE VIEWING AT https://www.myequals.net

THE AUSTRALIAN NATIONAL UNIVERSITY

THIS IS TO CERTIFY THAT

FOLLOWING THE COMPLETION OF
AN APPROVED PROGRAM OF STUDY

Subhransu Bhattacharjee

HAS BEEN AWARDED THE DEGREE OF

Bachelor of Engineering

WITH HONOURS

GIVEN UNDER THE SEAL OF THE AUSTRALIAN NATIONAL UNIVERSITY
ON THE THIRTEENTH DAY OF DECEMBER 2022

Chancellor

Vice - Chancellor

189512

VALID ONLY FOR ONLINE VIEWING AT https://www.myequals.net